



EDUCATION			
Program	Institution	CGA	Completion
M.Tech. in Data Science and Artificial Intelligence	Indian Institute of Technology, Madras	8.68	2026
B.Tech. in Computer Science & Eng.	R.V.R. & J.C. College of Engineering, Guntur	8.95	2024
Diploma in Data Science and Programming	Indian Institute of Technology, Madras	8.39	2024
XII (MPC)	Sri Chaitanya Junior College, Vijayawada	9.94	2020
X (SSC)	Ravindra Bharathi School, Tenali	10	2018
● Secured AIR 114 rank (99.71 percentile) in GATE'24 (Data Science and Artificial Intelligence)			

EXPERIENCE	
Data Scientist Intern <i>Walmart</i> <i>Bengaluru, India</i> <i>Aug'25 – present</i> <i>M.Tech Project</i>	● Building an end-to-end, modular Agentic AI Annotator to automate vendor manual operations (item-pair labelling, image validation). ● Integrating finetuned multimodal LLMs, LLM Gateway and agent orchestration with human-in-the-loop fallback and explainability logs.
Research & Development Intern <i>Samsung R&D</i> <i>Noida, India</i> <i>May'25 – Jul'25</i> <i>(PPO Offered)</i>	● Part of the Language AI Team, worked on User Training Persona project to model user engagement, app preferences using mobile usage data. ● Developed a pipeline for Continuous Pre-training and Fine-tuning of small language models to recommend similar apps based on user behavior and age group. ● Leveraging Hugging Face's TRL library with Proximal Policy Optimization (PPO), crafting a custom reward function on top of finetuned model to iteratively adjust network weights for improved recommendation quality.
Research Intern <i>MI Lab, IIT Madras</i> <i>Prof. Rohit Batra</i> <i>Feb'24 – Jul'24</i>	● Engaged in a project using Large Language Models to extract materials data from scientific literature. ● Developed a Data Extraction Pipeline with Elsevier Text Mining API and Beautiful Soup to convert unstructured articles into structured formats. ● Applied prompt engineering and Retrieval Augmented Generation (RAG) techniques using Langchain and FAISS on GPT-4 and Llama3 for extracting alloy-related data. ● Improved F1 score by 25%, achieving 97% with GPT-4 at 15 cents per paper; GPT-4 Omni achieved 95% at 0.3 cents per paper, using our evaluation metrics and criteria.
Frontend Developer Intern <i>Acqux</i> <i>Oct'23 – Jan'24</i>	● Worked at a startup founded by IIT Delhi alumni, collaborating closely with the team and gaining hands-on experience with project workflows and agile methodologies. ● Developed user interfaces with Vue.js and Bootstrap, implementing Figma designs into components. ● Integrated front-end with Django backend via APIs and used Pinia for state management.

PUBLICATION/CONFERENCE
Development and Analysis of Largest, Accurate and Comprehensive Multicomponent (high entropy) Alloy Database using Large Language Models - <u>Accepted</u>, <i>AI4X Singapore Conference, July 2025</i>
● Developed an LLM-based pipeline to extract alloy compositions, processing conditions, characterization methods, and physical/mechanical properties from text and tables across >10,000 articles, producing the largest publicly available multicomponent (HEA) alloy dataset. ● The pipeline (prompt engineering, RAG with GPT-4o, Llama-3, GPT-3.5) achieved 0.90 F1 on textual extraction and 0.95 F1 on tabular extraction, manuscript being prepared for journal submission.

PROJECTS	
Qwen2-VL LaTeX Finetuning	- Fine-tuned the Qwen2-VL model on a LaTeX OCR dataset to convert equation images into LaTeX code. - Developed an image-text pipeline leveraging PyTorch and UnsLoth for training and inference. - Tech stack: PyTorch, UnsLoth, Qwen2-VL, LaTeX OCR Dataset (Hugging face).
Seq2Seq Transliteration <i>Prof. Mitesh Kapra</i>	- Performed English to Telugu Transliteration using RNN/LSTM/GRU based Encoder-Decoder architectures on Dakshina dataset. Implemented Attention Mechanism to improve test and validation accuracy. Attained 57% validation accurcay and 52% test accuracy. - Tech stack: PyTorch, NumPy, pandas, Jupyter, WandB, Dakshina dataset.
Lymphoblastic Leukemia Classifier MLOps Project <i>Prof. Sudharsan</i>	- Developed an end-to-end MLOps pipeline to detect Acute Lymphoblastic Leukemia from white-blood-cell microscopy images, integrated FastAPI, Streamlit, Airflow, MLflow, Prometheus, Grafana and Docker for data ingestion, training, monitoring and production deployment. - Tech Stack: Python, FastAPI, Streamlit, Apache Airflow, MLflow, Prometheus, Grafana, Docker.
Hierarchical RL Taxi Driver Agent <i>Prof. Balaraman Ravindran</i>	- Developed a hierarchical reinforcement-learning agent for the Taxi domain, implementing Semi-Markov Decision Process (SMDP) based Q-learning and an Intra-Option Learning agent to master passenger pickup/drop-off tasks. - Leveraged options framework to make the agent learn fast and convergence was reached around 1000 episodes. - Tech stack: Python, Gymnasium, WandB.
Voice Analyzer Analyzes Speech Patterns	- Implemented speech transcription, translation, speech history storage, and word frequency analysis. - Identified unique phrases, integrated speech similarity detection among users and hosted the app. - Tech stack: Reactjs, Nodejs, MongoDB, Google Translate API.

COURSEWORK & SKILLS		
Deep Learning	Reinforcement Learning	Modern Computer Vision
Foundations of Machine Learning	Natural Language Processing	GPU Programming
MLOps Lab	Artificial Intelligence	Tools in Data Science
Skills	Programming Python, C++, Java, C Machine Learning Tensorflow, Pytorch, Langchain, Optuna, Mlflow, wandb Frameworks Flask, Vuejs, Nodejs, Reactjs, Bootstrap Database Management Oracle SQL, Redis, Vector Databases Cloud/Platforms Git, Github, Docker, Linux	

ACHIEVEMENTS	
Co-curricular/Academic	- Acknowledged among the top nine teams in our college Ideathon for the "Street Light Fault Detection Management System" and awarded mementos - Sep'23 - Ranked among the top ten in the 48-hour Skillathon in Erode, Tamil Nadu, besting 63 teams - Oct'23
POSITION OF RESPONSIBILITY	
Placement Coordinator July'24 - present	- Inviting and coordinating with companies to create placement opportunities, managing schedules, and overseeing the entire placement process for our department. - Handled companies like SAP Labs, Quantitative Brokers, Augnito AI ensuring the smooth execution of interviews and related activities.
Event Organizer ACM Student Chapter Oct'22 - Dec'23	- Organized coding events and contests to engage students across college. - Mobilized a team of volunteers and managed 1240 contestants for Medhanvesh event.